MIL-T-38328B
22 July 1981
SUPERSEDING
MIL-T-38328A(USAF)
6 December 1971

MILITARY SPECIFICATION

TAPE, TEXTILE, NYLON, AROMATIC, NONMELTING, REINFORCING

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes requirements for nine types of aromatic polyamide reinforcing tape.

1.2 Classification. The tape shall be of the following types, as specified (see 6.2):

Type I	Unaged breaking strength of 400 lbs.
Type II	Unaged breaking strength of 500 lbs.
Type III	Unaged breaking strength of 1,000 lbs.
Type IV	Unaged breaking strength of 1,500 lbs.
Type V	Unaged breaking strength of 525 lbs.
Type VI	Unaged breaking strength of 350 lbs.
Type VII	Unaged breaking strength of 1,500 lbs.
Type VIII	Unaged breaking strength of 2,000 lbs.
Type IX	Unaged breaking strength of 3,000 lbs.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-W-43334 Webbing and Tape, Textile, Packaging and Packing of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to the Aeronautical Systems Division, ENESS, WPAFB, Ohio by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 8315

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.

STANDARDS

FEDERAL

FED-STD-191	Textile Test Methods
FED-STD-595	Colors

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

Rules and Regulations Under the Textile Fiber Products Identification Act

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580.)

3. REQUIREMENTS

3.1 Yarn. The yarn shall be a nonmelting, aromatic, polyamide. The yarn shall not begin to char at a temperature less than 675°F when tested as specified in 4.2.1.

3.2 Weave

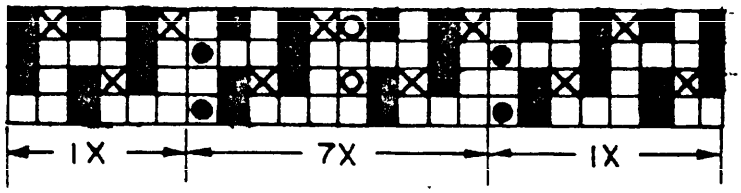
3.2.1 Types I and V. The weave of types I and V shall be plain (1-up and 1-down).

3.2.2 Types II and VII. The weave of types II and VII shall be 2-up and 2-down herringbone twill with 1 reversal in the center.

3.2.3 Types III and IV. The weave of types III and IV shall be composed of 2 ground warps (face and back), 1 binder warp, and 1 filling. The face warp shall weave plain with the picks that show on the face and the back warp shall weave plain, with the picks that show on the back. The binder warp shall weave plain throughout.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.

3.2.4 Type VI. The weave of type VI shall be as shown in Figure 1.



= Face warp - 2 ends as 1

= Back warp - 2 ends as 1

= Stuffer warp - 4 ends as 1

= Binder warp - 2 ends as 1

Figure 1. Weave diagram - Type VI.

3.2.5 Types VIII and IX. The weave of types VIII and IX shall be 2 up and 2 down right hand twill.

3.3 Color. Unless otherwise specified (see 6.2), the color of the tape shall be Sage Green, color 1565 in accordance with FED-STD-595. The color shall be obtained by the use of melt spun solution dyed yarn.

3.4 Physical properties. The physical properties of the tape shall be as specified in Table I, when tested as specified in 4.3.

3.5 Age. The tape shall be not more than 2 years old from the date of manufacture to the date of delivery.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

* 6.8 Changes from previous issue. The margins of this specification are

TABLE I. Physical properties.

REQUIREMENT	TYPE I	TYPE II	TYPE III	TYPE IV	TYPE V	TYPE VI	TYPE VII	TYPE VII	TYPE IX
Thickness, inch (max)	0.025	0.030	0.040	0.040	0.025	0.037	0.040	0.040	-
Width, inches	3/4 + 1/32	9/16 + 1/32	1 + 1/32	1-1/2 + 1/32	1 + 1/32	5/16 + 1/32	2 + 1/16	2 + 1/16	2 + 1/16
Weight, oz/yd (max)	0.25	0.25	0.50	0.75	0.30		0.95	1.00	1.06
Breaking strength (lb)									
Unaged (min)	400	500	1,000	1,500	525	350	1,500	2,000	3,000
Aged (% of unaged)(min)	85	85	85	85	85	85	85	85	85
Elongation, % (min)	18	12	15	18	15				
Warp yarns:									
Face & back (min)	210	60	241	363	276	138 (2 as 1)	376	512	260
Denier and ply	200/1	200/4	200/2	200/2	200/1	200/1	200/2	200/2	1200/1
Twist (min)									
Ply	-	2.5	2.5	2.5	2.5	-	-	-	-
Singles	5.0	-	-	-	-	-	-	-	-

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.

TABLE I. Physical properties (Contd)

REQUIREMENT	TYPE I	TYPE II	TYPE III	TYPE IV	TYPE V	TYPE VI	TYPE VII	TYPE VII	TYPE IX
Binder yarn (min)	-	-	18	26	-	16	-	-	-
Denier and ply	-	-	200/2	200/2	-	200/1	-	-	-
Twist (min)									
Ply	-	-	2.5	2.5	-	-	-	-	-
Singles	-	-	-	-	3.0	3.0	-	-	-
Stuffer (min)	-	-	-	-	-	28 (4 as 1)	-	-	-
Denier and ply	-	-	-	-	-	200/1	-	-	-
Twist (min)	-	-	-	-	-	-	-	-	-
Singles	-	-	-	-	-	3.0	-	-	-
Filling yarns:									
Denier and ply	200/2	200/2	200/1	200/2	200/2	200/1	200/4	200/2	200/2
Twist (min)									
Ply	2.5	2.5	2.5	2.5	2.5	-	-	-	-
Singles	-	-	-	-	-	2.0	-	-	-
Picks per inch (min)	27	32	44	44	27	68 (2 as 1)	32	35	24

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.

3.6 Length and put-up. Unless otherwise specified, tape shall be put up on rolls not less than 90 or more than 110 yards. Each roll shall contain not more than 3 pieces, and no piece shall be less than 10 yards in length.

3.7 Identification of product. Each roll shall have a ticket (identification tag) attached with not finer than 5-ply cotton string doubled to not less than 6 inches. The ticket shall be made of paperboard not less than 0.020 inch in thickness and the color shall be manila or light in intensity to permit easy reading of printed, stamped, or typed markings. Handwritten entries shall be prohibited. The ticket shall have clipped corners at the end where a reinforcing patch (with or without a metal eyelet) is firmly affixed for attaching the tying strings. The ticket shall be legibly marked with water-insoluble ink with the following information:

Nomenclature
Number and date of this specification
Type and width
Yardage
Manufacturer's identification number
Contract number; name of contractor
Stock number
Date of manufacture (month and year)
Name of manufacturer (if not the same as contractor).

3.8 Fiber identification. Each roll of tape shall be labeled, ticketed, or invoiced for fiber content in accordance with the Textile Fiber Products Identification Act, effective 3 March 1960.

3.9 Workmanship. The tape shall be clean, evenly woven, and shall conform to the quality and grade of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of all inspection requirements as specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.

4.2 Inspection for acceptance. Except where otherwise specified, inspection shall be in accordance with MIL-STD-105.

4.2.1 Inspection of component. The component listed in Table II shall be tested for the characteristics specified in accordance with the referenced test methods of FED-STD-191, except where otherwise specified. The sample unit shall be one cone, tube, or spool and the lot size shall be expressed in units of one cone, tube, or spool each. The sample size shall be in accordance with inspection level S-1 of MIL-STD-105 and the acceptable quality level for each characteristic shall be 4.0 test failures per 100 units.

Table II. Testing of component.

Component	Characteristic	Requirement reference	Test method of FED-STD-191 or test reference
Yarn	Denier Ply Twist Carbonization	Table I Table I Table I 3.1	4021 Visual 1/ 4054 1534 2/

1/ One determination per sample unit and results reported as pass or fail.

2/ The supplier's certificate of compliance showing that the yarn identified in 6.3 was used may be submitted in lieu of test method.

4.2.2 Examination of the end item. Defects found during the examination shall be classified in accordance with 4.2.2.1, 4.2.2.2, and 4.2.2.3.

4.2.2.1 Yard-by-yard examination. The required yardage of each roll shall be inspected and visual defects classified as listed in table III. The defects found shall be counted regardless of their proximity to each other, except where two or more defects represent a single local condition of the tape, in which case only the more serious defect shall be counted. A continuous defect shall be counted as one defect or each warpwise yard, or fraction thereof,

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

in which it occurs. The unit of product for this examination shall be 1 linear yard. The sample size shall be in accordance with inspection level III of MIL-STD-105. The acceptable quality level shall be 0.40 major defects and 1.50 total defects per 100 units (yards). The lot size shall be expressed in units of 1 linear yard each. An approximate equal number of yards shall be examined from each roll selected. The number of rolls from which the sample is to be selected shall be in accordance with Table IV of the specification.

Table III. Classification of defects.

Defect	Description 1/	Major	Minor
Abrasion marks	Resulting in rupture of yarns or in nap sufficient to obscure the identity of any yarns exceeding 10 percent of width or 1 inch in length.	X	
Broken or missing end	Two or more regardless of length or a single end exceeding 6 inches in length.	X	
	Single end under inches but exceeding 1/4 inch.		X
Broken or missing pick	Two or more regardless of extent.	X	
	One or single.		X
Coarse or light filling bar	Resulting in clearly noticeable difference in stiffness or thickness of tape and extending for more than 1/4 inch in the length direction.	X	
	Resulting in clearly noticeable difference in stiffness or thickness of tape and extending for 1/4 inch or less in the length direction.		X
Cut, hole, or tear	Any cut, hole, or tear	X	

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

Table III. Classification of defects. (Continued)

Defect	Description 1/	Major	Minor
Drop-ply	Clearly noticeable, on more than 2 ends within same length and extending over 9 linear inches or more.	X	
	Clearly noticeable, on 1 or 2 ends within same 9 linear inches or more.		X
Edges	Frayed, slack, or otherwise poorly constructed and exceeding 1/4 inch in length.	X	
Fine or light filling bar, light place	Clearly noticeable.	X	
Floats or skips	Multiple, 1/2 inch or more in warp and filling directions.	X	
	Single float or skip over 1 inch.	X	
	Multiple, less 1/2 inch in combined warp and filling directions.		X
	Single float or skip over 1/2 inch if in the warp.		X
	Single float or skip over 1/4 inch but not exceeding 1 inch if in the filling.		X
Hitchback crack	Clearly noticeable opening between adjoining picks, or warpwise tension area over part of the width resulting in noticeable light and heavy places.		X
Jerked-in filling, slough-off, slug	More than twice the thickness of the normal yarn.		X
Kinks	More than 3 in any 9 linear inches.	X	

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

Table III. Classification of defects. (Continued)

Defect	Description 1/	Major	Minor
Knots	More than 2 knots in any 9 linear inches.	X	
	Single knot with untrimmed ends extending more than 1/16 inch from surface of tape.	X	
Mispick, double pick	Two or more across the full width.	X	
	Single across the full width.		X
Slack end	Two or more in the same length, jerked in between picks, or forming clearly noticeable loops on the surface of tape.	X	
	Single jerked in between picks, or forming clearly noticeable loops on surface of tape.		X
Slub (slug), or gout	More than twice the thickness of the yarn (or ply, if plied).		X
Smash	Any Smash.	X	
Spot, stain, or streak	Any clearly noticeable dirt, rust, grease, oil spot, stain, or streak.		X

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

Table III. Classification of defects. (Continued)

Defect	Description 1/	Major	Minor
Tight end	Clearly noticeable.	X	
Tight pick or tight filling	Resulting in rolling of tape.	X	
Twist or distortion	Twisted or distorted. Will not lay flat upon application of manual pressure.		X
Weak, soft, or tender spot	Any weak, soft, or tender spot.	X	
Width	Beyond specified tolerance.		X
Wrong draw	Extending for more than 9 inches.		X

1/ The term clearly noticeable shall be interpreted to mean visible at normal inspection distance (approximately 3 feet).

4.2.2.2 Overall examination. Each defect listed herein shall be counted no more than once in each roll examined. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table IV. The lot size shall be expressed in units of 1 linear yard each.

Defects

Objectional odor.

Overall uncleanness

Uneven weaving throughout.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

4.2.2.3 Examination for length.

4.2.2.3.1 Individual roll. Each roll in the sample shall be examined for the defects listed herein. The sample unit for this examination shall be one roll. The sample size and acceptance number shall be as shown in table IV. The lot size shall be expressed in units of 1 linear yard each.

Defects

Gross length less than specified minimum length or more than specified maximum length.

Gross length more than 2 yards less than gross length marked on piece ticket.

Any piece less than the allowable minimum length of piece.

Any roll containing more than the allowable number of pieces.

4.2.2.3.2 Total yardage. The lot shall be unacceptable if the total of the actual gross lengths of rolls in the sample is less than the total of the gross lengths marked on roll tickets.

4.2.3 Examination for compliance with Texitle Fiber Products Identification Act. During the examination of individual roll for length, each roll in the sample shall be examined for conformance to the Textile Fiber Products Identification Act. Each roll not labeled in accordance with this act shall be a defect. The lot shall be unacceptable if two or more of these defects occur.

Table IV. Sample size.

Lot size in yards	Sample size in rolls	Maximum number of defects acceptable in sample
Up to 1,200 1/	5	0
1,201 to 3,200	7	0
3.201 to 10,000	10	0
10,001 to 35,000	15	0
35,001 to 150,000	25	1
150,001 and over	35	1

1/ If a lot contains fewer than 5 rolls, each roll in the lot shall be examined.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

4.2.4 Examination of preparation for delivery requirements. An examination shall be made in accordance with the provisions of MIL-W-43334 to determine that packaging, packing, and marking comply with the Section 5 requirements.

Examination	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, of improper size, location, sequence or method of application.
Materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as: incomplete closure of case liners, container flaps, loose strapping, inadequate stapling. Bulging or distortion of containers.
Weight	New weight exceeds requirements.

4.3 Testing of the end item. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table V shall be followed. The physical and chemical values specified in Section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the application test methods. The sample size shall be as follows:

Lot size (yards)	Sample size
800 or less	2
801 to 10,000	3
10,001 and over	5

The lot shall be unacceptable if one or more units fail to meet any requirements specified. The lot size shall be expressed in units of 1 linear yard. The sample unit for testing shall be 10 yards.

4.3.1 Breaking strength after aging. The size of the specimens for oven aging shall be the same as specified in 4.3. The specimens shall be placed in an oven at $500^{\circ} \pm 10^{\circ}$F for 4 hours. Upon removal, the specimens shall be conditioned at $70^{\circ} \pm 2^{\circ}$F and 65 $\pm$ 2 percent relative humidity for 4 hours and tested for breaking strength. The loss in breaking strength due to the aging treatment shall be reported as percent loss from the unaged specimens.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

Table V. Test methods.

Characteristics	Requirement reference	Test method of Fed Std No. 191 test reference	Number of determinations per indiv. sample unit 1/	Results reported as 1/
Texture				
Warp yarns	Table I	5050 2/	3	Nearest whole number
Binder yarns	Table I	5050 2/	3	Nearest whole number
Picks per inch	Table I	5050	-	-
Thickness	Table I	5030 3/	-	-
Width (inches)	Table I	5020	5	Nearest 1/32"
Weight	Table I	5040	-	-
Breaking strength				
Unaged 4/	Table I	4108	5	Nearest 1.0 lbs.
Aged	Table I	4108	5	Nearest 1.0% loss
Elongation 5/	Table I	4108	5	Nearest 1.0%
Weave	3.2	Visual	1	Pass or fail

1/ When the data in the "number of determinations" and "results reported as" columns are not specified in the table, they shall be as required by the referenced test method.

2/ Determination shall be made on full width of tape.

3/ Except that a 6-oz total load shall be applied and the presser foot shall be 3/8 inch.

4/ Type IV - No load rate of jaw separation at 4 inches/minute.

5/ In lieu of an elongation recording device on the testing machine, the elongation determination shall be in accordance with the following:

a. Minimum distance of 5 inches between bench marks on test specimen.

b. Maximum extension at 90 percent of breaking load.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.

5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Levels A and C. Tape, put up as specified, shall be packaged in accordance with the applicable requirements of MIL-W-43334.

5.2 Packing. Packing shall be level A, B, or C as specified (See 6.2).

5.2.1 Levels A, B, and C. Tape shall be packed in accordance with the applicable requirements of MIL-W-43334.

5.3 Marking. In addition to any special marking required by the contract or order, shipments shall be marked in accordance with MIL-W-43334.

6. NOTES

6.1 Intended use. The tape covered by this specification is intended for binding and reinforcing applications in parachute packs and similar purposes and for equipage.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Type and width desired (see table I).
c. Quantity.
d. Length and put-up if other than specified in 3.6.
e. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).
f. Color (see 3.3).

6.3 Nomex (formerly called HT-1), manufactured by E. I. duPont deNemours and Co., Inc. is an acceptable yarn.

Custodian:

Navy - SA
Air Force - 11

Preparing activity:
Air Force - 11

Project Number: 8315-0291

Reviewer activity:

Air Force - 99
DLA - CT

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2 REMARKS

SUBMITTED BY *(Printed or typed name and address – Optional)*

TELEPHONE NO.

DATE

DD FORM 1 OCT 76 1426 PREVIOUS EDITION WILL BE USED.

Source: https://assist.dla.mil -- Downloaded: 202610-21T12:23Z
Check the source to verify that this is the currentversion before use.

FOLD

DEPARTMENT OF THE AIR FORCE

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

NO POSTAGE
NECESSARY
IF MAILED
IN THE
UNITED STATES

BUSINESS REPLY MAIL

FIRST CLASS PERMIT NO. 73236 WASHINGTON D. C.

POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE AIR FORCE

ASD/ENESS
Wright-Patterson AFB, O 45433

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T12:23Z
Check the source to verify that this is the current version before use.